Journal of Applied Physics

RESEARCH ARTICLE | SEPTEMBER 01 2022

Ion trap frequency measurement from fluorescence dynamics

K. Saito ; R. Saito ; T. Mukaiyama

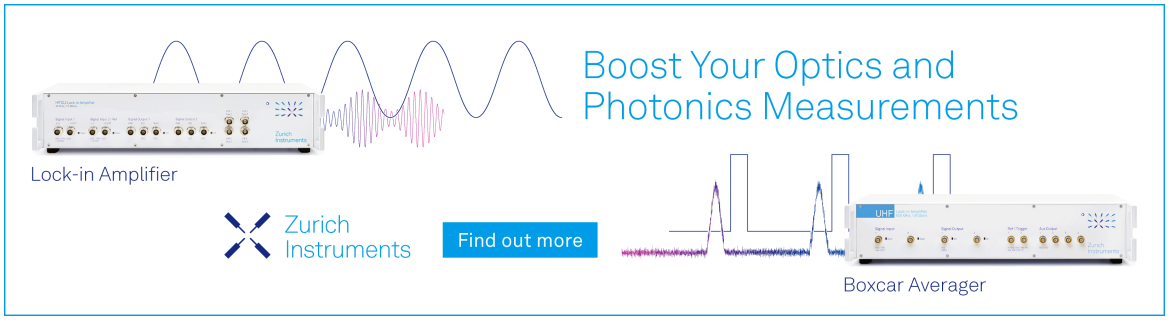
Check for updates

J. Appl. Phys. 132, 094401 (2022)
https://doi.org/10.1063/5.0100007

CHORUS

View Online
Export Citation
CrossMark

05 March 2024 09:22:54



AIP Publishing

Journal of Applied Physics | METHOD | scitation.org/journal/jap

Ion trap frequency measurement from fluorescence dynamics

Cite as: J. Appl. Phys. **132**, 094401 (2022); doi: 10.1063/5.0100007
Submitted: 20 May 2022 · Accepted: 9 August 2022 ·
Published Online: 1 September 2022

View Online Export Citation CrossMark

K. Saito,[1,a)] R. Saito,[1,2] and T. Mukaiyama[1,2,b)]

AFFILIATIONS

[1]Graduate School of Engineering Science, Osaka University, 1-3 Machikaneyama, Toyonaka, Osaka 560-8531, Japan
[2]Quantum Information and Quantum Biology Division, Institute for Open and Transdisciplinary Research Institute Osaka University, Osaka 560-8351, Japan

[a)]u463777j@ecs.osaka-u.ac.jp
[b)]**Author to whom correspondence should be addressed:** muka@ee.es.osaka-u.ac.jp

ABSTRACT

In this paper, we propose and demonstrate a new method for measuring the trap frequency from the fluorescence dynamics of an ion, initiated using fast trap-center displacement. Here, we observe the coherently excited motion of an ion in a harmonic trap using time-resolved fluorescence detection. Furthermore, Fourier analysis of the ion fluorescence signal was used to determine the trap frequency of an ion with high precision and reproducibility. The results show that our proposed method can be an alternative way to determine trap frequencies in an ion trap experiment.

© 2022 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (http://creativecommons.org/licenses/by/4.0/). https://doi.org/10.1063/5.0100007

05 March 2024 09:22:54

I. INTRODUCTION

Laser-cooled ions stored in an ion trap are a valuable platform for developing devices that take advantage of quantum phenomena. Trapped ions in a vacuum are considered well-isolated systems with many applications, particularly in atomic frequency standards,[1–3] quantum information,[4,5] and quantum sensing.[6,7] Ions in Paul traps are dynamically trapped by RF electric fields but can be regarded as trapped in a harmonic potential when time-averaged. The trap frequency characterizes various experimental parameters, and the importance of accurately measuring it has been widely recognized. Several methods for performing this task have been proposed and implemented to date. One method is based on motional sideband spectroscopy.[8] Spectroscopy of a narrow transition can be used to observe blue and red sideband components, whose frequencies include the trap frequency information. Another widely used scheme involves the observation of heating by resonant excitation.[9–12] This scheme is simple and conventional since it can measure the trap frequency by adding an oscillating voltage onto one of the trap electrodes and observing the decrease in ion fluorescence. In this scheme, the ion needs to be excited such that its fluorescence decreases by a detectable amount. However, the ion responds sensitively to the modulation, and, therefore, such intense heating of the ion evokes the potential randomness of the excitation and causes the shift and instability in the observed trap frequency. This instability limits the precision of the trap frequency in the resonant modulation scheme.

In this paper, we present a method for measuring trap frequency from the fluorescence dynamics of an ion, using fast trap-center displacement.[13,14] The sudden trap-center displacement creates a coherent excitation of the ion motion, which coherently modulates the ion fluorescence. The time-resolved detection of ion fluorescence is used to coherently observe the modulation phase, and therefore, the trap frequency of an ion can be determined. Contrary to schemes with resonant driving such as sideband spectroscopy or resonant excitation, the proposed scheme is unique in that we need not know the trap frequency in advance.

II. EXPERIMENT SETUP AND SCHEME

In this study, a $^{174}Yb^{+}$ ion was Doppler-cooled and trapped with a conventional linear Paul trap with nine electrodes consisting of two RF, two RF ground (GND), four endcap (END) electrodes, and a compensation electrode (COMP).[15,16] The ion was radially

© Author(s) 2022

confined by the RF field and axially confined by the static field along the symmetry axis of the trap. Under our trapping conditions, the RF field was applied using a helical resonator with a peak-to-peak voltage of 1040 V_{p-p} at a resonance frequency of $\Omega = 2\pi \times 23.9$ MHz. The radial trap frequencies were $\omega_{rad} = 2\pi \times (1.76\,\text{MHz}, 1.55\,\text{MHz})$. We applied 28.4 V to the END electrodes, and the axial trap frequency was $\omega_{ax} = 2\pi \times 0.77$ MHz. We deliberately ramped up the voltage applied to one of the GND electrodes by 0.3 V to displace the trap center by 3.7 μm and returned the voltage to the original value after 0.4 μs. Because 0.4 μs does not match the oscillation period of the ion along any trap axes, the ion motion was excited after the voltage pulse. Furthermore, a voltage of 0.3 V and a delay time of 0.4 μs were used to determine the amplitude of the ion oscillation. However, the measured trap frequencies are unaffected by the choice of these values as long as the oscillation amplitude of the ion is small enough to avoid deformation of the fluorescence signal.[14] The rise time of the voltage pulse was 5 ns, which is much shorter than the time scale of any dynamics investigated. The fluorescence from the ion was collected by an objective lens with a numerical aperture of 0.28, enabling the collection of ion fluorescence with an efficiency of 0.6%. The fluorescence collected from the ion reached an electron-multiplying charge-coupled device (EM-CCD) camera and a photomultiplier tube (PMT). To observe the fluorescence dynamics, the number and arrival time of the fluorescence photons at the PMT were recorded using a multichannel scaler within 100 ns.

Figure 1(a) shows our scheme for exciting ion motion. When the trap center of the ion is displaced, the ion begins to oscillate [middle in Fig. 1(a)]. After holding the situation for a duration of approximately half-integer times the trap period, we turned the trap center back to its original position. The holding time was chosen to efficiently excite the ion motion.[14] Simultaneously, we irradiated the ion with a cooling laser to detect the fluorescence [bottom in Fig. 1(a)]. The Doppler shift of the cooling laser ($\Delta f = kv$, where k and v are the wavenumbers of the laser and velocity of the ion, respectively) was modulated by the trap frequency due to the ion oscillation; therefore, the ion fluorescence was also modulated with the same frequency [Fig. 1(b)].[17–19] By analyzing the modulation frequencies of ion fluorescence, we precisely determined the trap frequencies.

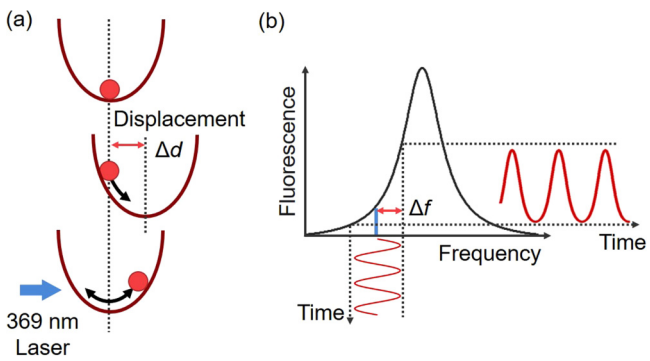


FIG. 1. (a) Schematic of our scheme for exciting ion motion. The trap center is displaced by Δd, and returned to the original position after a hold time. Then, the fluorescence of the ion oscillation is detected by irradiating the cooling laser to the ion. (b) Schematic of the change in fluorescence due to the oscillatory motion of the ion. The blue line indicates the frequency of the irradiated cooling laser. The red wave at the bottom shows the frequency of the cooling laser in the restframe of the ion, that is, Δf, modulated by the Doppler shift due to the oscillation. The fluorescence of the ion oscillates, as shown by the red wave on the right.

III. RESULT

Figure 2(a) shows the typical fluorescence signal obtained using PMT and time-resolved by the multichannel scaler integrated 10 000 times. The initial part of the fluorescence dynamics shows a relatively large modulation amplitude, reflecting the ion oscillation in the trap. After 0.4 ms of holding, the modulation of the fluorescence signal was halved, and vanished in 1 ms due to the Doppler cooling of the ion.

Figures 2(b) and 2(c) show the Fast Fourier Transforms (FFT) of the fluorescence signal from 0 to 1 ms [Fig. 2(b)] and from 2 to 3 ms [Fig. 2(c)], respectively. With a time bin of 100 ns and 1 ms of fluorescence dynamics measurements, our Fourier analysis had a frequency resolution of 1 kHz and a measurement frequency range of approximately 5 MHz. Three prominent peaks were observed in Fig. 2(b), indicated by rad1, rad2, and ax, corresponding to the trap frequencies in the radial and axial directions, respectively. In this setup, the cooling light had wavevector components in both the axial and radial directions. Therefore, the fluorescence signal included the modulation due to the ion motion along the radial

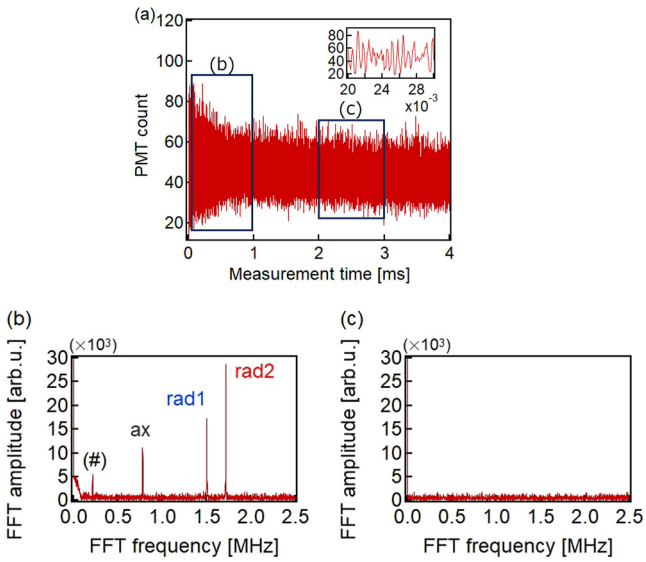


FIG. 2. (a) Typical time-resolved fluorescence signal obtained using PMT. The vertical axis represents the number of photons detected by PMT within 100 ns and integrated 10 000 times. The inset shows fluorescence from 20 to 30 μs. (b) and (c) show the Fourier transform of the plots from 0 to 1 ms (b), and from 2 to 3 ms (c), respectively.

05 March 2024 09:22:54

© Author(s) 2022

and axial directions. However, the excitation of the ion motion was applied only in the radial direction. Therefore, the feasibility of measuring the axial trap frequency remains unclear. We detected the Fourier peak corresponding to the axial motion, presumably due to imperfection in the excitation direction. The peak represented by (#) indicates the difference between the two radial trap frequencies. The inset of Fig. 2(a) shows the enlarged signal from 20 to 30 μs, which shows the beat between two radial frequencies. Conversely, no peak was observed in the spectrum shown in Fig. 2(c). Although the fluorescence signal had finite fluctuations, it contained no specific frequency components because the fluorescence fluctuation was caused by the random motion of the ion after Doppler cooling.

Figure 3(a) shows the measured trap frequencies with the peak-to-peak RF voltage ranging between 580 V_{p-p} to 1100 V_{p-p}. The red and blue markers show the experimentally obtained trap frequencies in two radial directions, which show a linear dependence on the RF voltage. Black markers show the trap frequency in an axial direction, which is insensitive to the RF voltage. The RF voltage dependence indicates that the red and blue markers correspond to the radial directions, whereas the black markers correspond to the axial direction.

Figures 3(b) and 3(c) show ten-times repeated frequency measurements at the blue marker on the vertical dashed line in Fig. 3(a) using our method [Fig. 3(b)] and resonant modulation methods [Fig. 3(c)], respectively. The Fourier spectrum around the resonance frequency is shown in Fig. 3(b), and the modulation frequency was scanned from −30 to +70 kHz with respect to the resonance frequency in Fig. 3(c). The spectral width is narrower in each measurement using our proposed method than using the conventional method. The spectral width (full width at half maximum, FWHM) of a single dataset measured using our method was 2.5 kHz, determined by the transform limit of the duration of the fluorescence signal. Contrarily, the spectral width (FWHM) in the resonant modulation scheme was measured as 10.3 kHz, which is four times broader than the width obtained using the proposed scheme. Furthermore, the measurement by the resonant modulation scheme suffers from stochastic heating, causing the random drift of the modulation spectrum from shot to shot. This stochastic drift of the spectrum limits the accuracy of the trap frequency measurement with the resonant modulation scheme.

Added to the difference in the spectral widths of both methods, there is a practical difference in the reproducibility of trap frequency measurements. The poor reproducibility arises from the fact that the resonant modulation scheme excites the ion as much as the point where the time-integrated ion fluorescence decreases by a detectable amount. When the modulation frequency is scanned and hits the trap frequency, the ion is heated, and sometimes the resonance dip broadens due to the delay in the recooling of the ion. In principle, this issue can be avoided by choosing the appropriate modulation intensity and duration. However, in reality, the degree of ion heating changes with time. Therefore, the reproducibility of the trap frequency dip in the resonant modulation scheme is poor compared to that of the proposed ion excitation method owing to the fast trap-center displacement.

Figure 4 shows the long-term measurements of the trap frequencies along the two radial directions [Figs. 4(a) and 4(b)] and

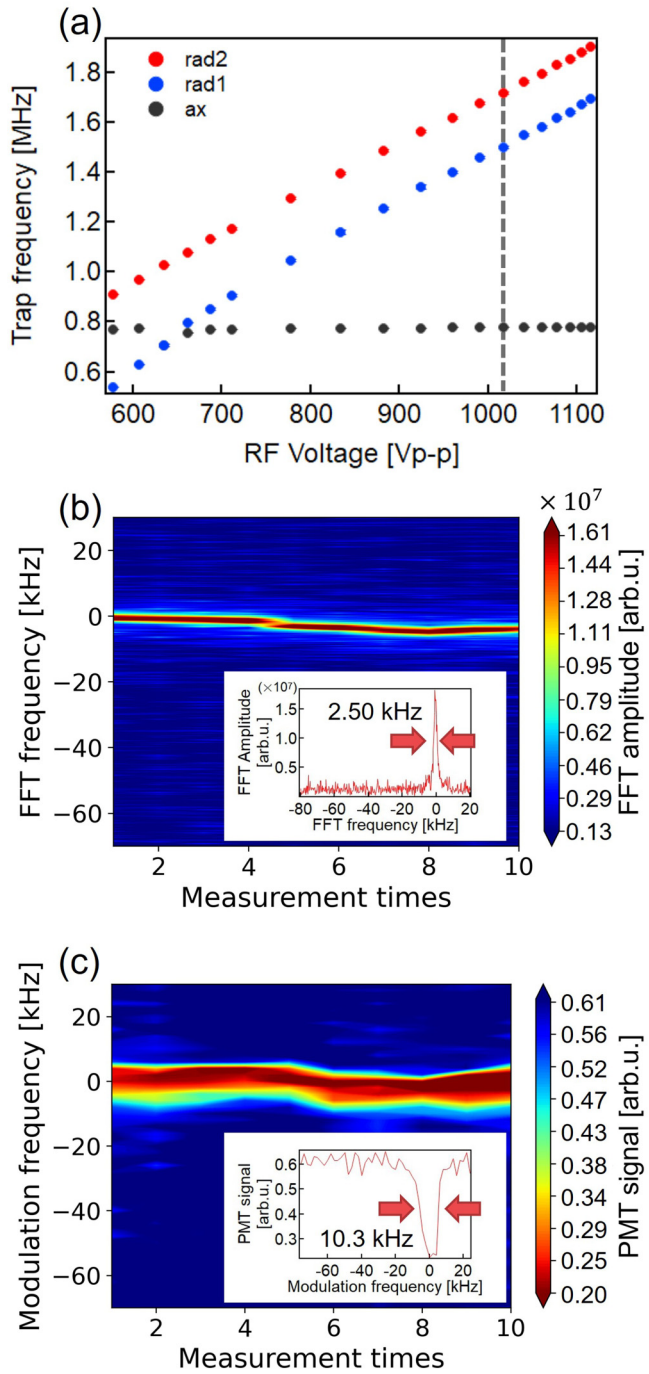


FIG. 3. (a) Trap frequencies vs applied RF voltage (peak-to-peak). Each set of data represents the spectral peak shown in Fig. 2(b). (b) represents ten measurements for the mode at the blue marker on the vertical dashed line in (a) using the proposed method. The inset in (b) shows one of the typical FFT spectra. (c) represents ten measurements for the same mode using the resonant excitation method. The inset in (c) shows one of the typical modulation spectra.

05 March 2024 09:22:54

© Author(s) 2022

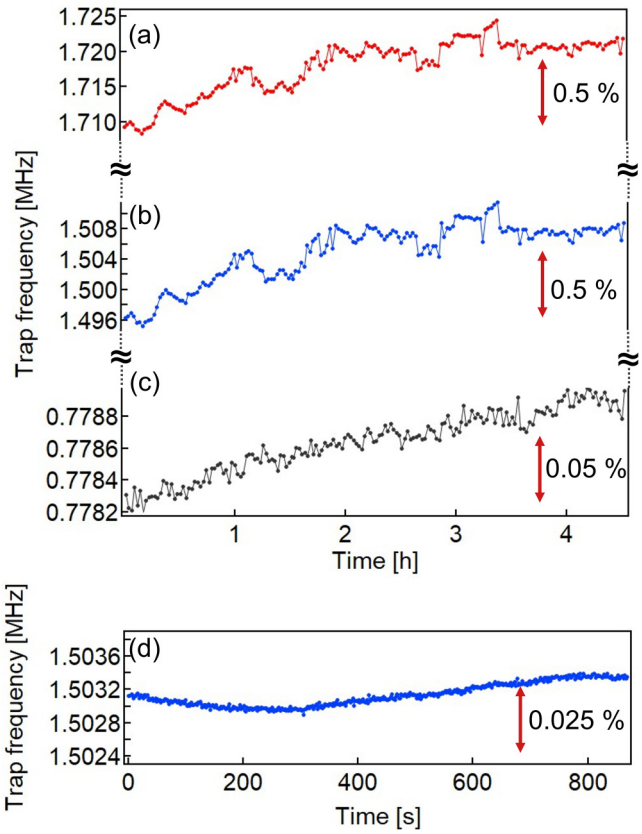


FIG. 4. Trap frequencies measured in every 92 s for 1 h are shown. (a) and (b) represent the trap frequency in the radial direction. The vertical arrow indicates the frequency fluctuation of 0.5%. (c) represents the trap frequency in the axial direction. The vertical arrow indicates the frequency fluctuation of 0.05%. (d) shows the measurement of one of the radial directions done every 1.8 s with the RF amplitude stabilization.

an axial [Fig. 4(c)] direction using our method, respectively. The trap frequencies were measured with a precision of 2 kHz with a measurement time of 92 s. The method was sufficiently precise to observe the variation in the trap frequency in the radial direction due to the change in the RF amplitude on the order of 1%, as shown in Figs. 4(a) and 4(b). From Fig. 4(c), the axial trap frequency varies similarly to the radial trap frequencies. This is because the oscillating RF field partly contributes to the axial confinement, and the variation in the RF amplitude causes a change in the axial trap frequency on the order of 0.1%. Figure 4(d) shows the measurement of one of the radial directions performed every 1.8 s with the RF amplitude stabilization.[20] It is seen that the trap frequency stability is improved, and the fluctuation of the trap frequency is measured to be within 0.01% in 870 s.

IV. CONCLUSION

In this paper, we developed a method for measuring trap frequency by observing the fluorescence dynamics of an ion after fast trap-center displacement. This method uses the idea that the time-resolved fluorescence signal reflects ion motion.[17,21] The method was compared with the widely used resonant modulation scheme and outperformed it in precision and reproducibility. The proposed scheme can be an alternative to other trap frequency measurement schemes in an ion-trap experiment.

Furthermore, FFT spectra around the overtone frequency may provide information on the anharmonicity of the trap, which is difficult to be determined in conventional schemes. In the current work, we do observe the overtone frequencies in the FFT spectra, but unfortunately, we remain unable to identify the source of the overtone as the trap anharmonicity. Further improvements in the resolution in the FFT analysis may enable us to identify the trap anharmonicity in our scheme.

ACKNOWLEDGMENTS

This work was supported by the JST-Mirai Program (Grant No. JPMJMI17A3) and the JST (Grant No. JPMJPF2015).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

K. Saito: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – original draft (equal). **R. Saito:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **T. Mukaiyama:** Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

05 March 2024 09:22:54

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1] N. Huntemann, C. Sanner, B. Lipphardt, C. Tamm, and E. Peik, "Single-ion atomic clock with 3×10^{-18} systematic uncertainty," Phys. Rev. Lett. **116**, 063001 (2016).
[2] S. M. Brewer, J.-S. Chen, A. M. Hankin, E. R. Clements, C.-W. Chou, D. J. Wineland, D. B. Hume, and D. R. Leibrandt, "$^{27}Al^{+}$ quantum-logic clock with a systematic uncertainty below 10^{-18}," Phys. Rev. Lett. **123**, 033201 (2019).
[3] A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, and P. O. Schmidt, "Optical atomic clocks," Rev. Mod. Phys. **87**, 637–701 (2015).
[4] R. Blatt and D. Wineland, "Entangled states of trapped atomic ions," Nature **453**, 1008–1015 (2008).
[5] C. Monroe and J. Kim, "Scaling the ion trap quantum processor," Science **339**, 1164–1169 (2013).
[6] W. Campbell and P. Hamilton, "Rotation sensing with trapped ions," J. Phys. B **50**, 064002 (2017).
[7] C. L. Degen, F. Reinhard, and P. Cappellaro, "Quantum sensing," Rev. Mod. Phys. **89**, 035002 (2017).
[8] D. Wineland, W. M. Itano, J. Bergquist, and R. G. Hulet, "Laser-cooling limits and single-ion spectroscopy," Phys. Rev. A **36**, 2220 (1987).

© Author(s) 2022

[9]F. Vedel and M. Vedel, "Nonlinear effects in the detection of stored ions," Phys. Rev. A **41**, 2348 (1990).
[10]R. Alheit, X. Chu, M. Hoefer, M. Holzki, G. Werth, and R. Blümel, "Nonlinear collective oscillations of an ion cloud in a Paul trap," Phys. Rev. A **56**, 4023 (1997).
[11]M. Razvi, X. Chu, R. Alheit, G. Werth, and R. Blümel, "Fractional frequency collective parametric resonances of an ion cloud in a Paul trap," Phys. Rev. A **58**, R34 (1998).
[12]X. Zhao, V. Ryjkov, and H. Schuessler, "Parametric excitations of trapped ions in a linear RF ion trap," Phys. Rev. A **66**, 063414 (2002).
[13]J. Alonso, F. M. Leupold, Z. Solèr, M. Fadel, M. Marinelli, B. C. Keitch, V. Negnevitsky, and J. Home, "Generation of large coherent states by bang–bang control of a trapped-ion oscillator," Nat. Commun. **7**, 1–8 (2016).
[14]R. Saito and T. Mukaiyama, "Generation of a single-ion large oscillator," Phys. Rev. A **104**, 053114 (2021).
[15]R. Saito, K. Saito, and T. Mukaiyama, "Measurement of ion displacement via RF power variation for excess micromotion compensation," J. Appl. Phys. **129**, 124302 (2021).
[16]A. Shinjo, M. Baba, K. Higashiyama, R. Saito, and T. Mukaiyama, "Three-dimensional matter-wave interferometry of a trapped single ion," Phys. Rev. Lett. **126**, 153604 (2021).
[17]M. J. Biercuk, H. Uys, J. W. Britton, A. P. VanDevender, and J. J. Bollinger, "Ultrasensitive detection of force and displacement using trapped ions," Nat. Nanotechnol. **5**, 646–650 (2010).
[18]T. B. Mitchell, J. Bollinger, X.-P. Huang, and W. M. Itano, "Doppler imaging of plasma modes in a Penning trap," Opt. Express **2**, 314–322 (1998).
[19]D. Berkeland, J. Miller, J. C. Bergquist, W. M. Itano, and D. J. Wineland, "Minimization of ion micromotion in a Paul trap," J. Appl. Phys. **83**, 5025–5033 (1998).
[20]K. G. Johnson, J. D. Wong-Campos, A. Restelli, K. A. Landsman, B. Neyenhuis, J. Mizrahi, and C. Monroe, "Active stabilization of ion trap radio-frequency potentials," Rev. Sci. Instrum. **87**, 053110 (2016).
[21]A. Poindron, J. Pedregosa-Gutierrez, C. Jouvet, M. Knoop, and C. Champenois, "Non-destructive detection of large molecules without mass limitation," J. Chem. Phys. **154**, 184203 (2021).

05 March 2024 09:22:54

© Author(s) 2022